

P17A4

Diagnostic information - PDK control unit

Electric pressure adjuster 4 (cooling flow)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 4 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 4 = 0 mA
- Momentary current of low-side pressure adjuster 4 $\geq$ 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side electric pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift

rod

- ◆ The maximum system pressure is increased to prevent activation of the emergency stop function
- ◆ Pressure-sensor plausibility check is switched off
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

P17A5

Diagnostic information - PDK control unit

Electric pressure adjuster 5 (synchro selection 6, R, 3, 5)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 5 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 5 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 5 = 0 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop

function

- ◆ The last engaged gear is held

P17A6

Diagnostic information - PDK control unit

Electric pressure adjuster 6 (synchro selection 4, 2, 1, 7)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 6 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function

♦ Last engaged gear is held

Diagnosis information PDK control unit

Electrical pressure adjuster 7

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 7 faulty

Fault setting condition(s)

- Momentary current of low-side electrical pressure adjuster 6 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side electrical pressure adjuster 6 = 850 mA
- Momentary current of low-side electrical pressure adjuster 6 = 0 mA

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Controlled transverse-differential lock is deactivated
- ♦ Hydraulic control 4 is switched off
- ♦ Low-side electrical pressure adjuster 7 is switched off
- ♦ Shiftlock solenoid is deactivated

P17A9

Diagnostic information - PDK control unit

Electric pressure adjuster 1 (clutch 1)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 1 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 1 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 1 = 0 mA
- Momentary current of low-side pressure adjuster 1 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 1 is switched off
- ♦ Low-side electric pressure adjuster 1 is switched off
- ♦ Only 2nd, 4th and 6th gears are possible
- ♦ No R gear
- ♦ Moving off in 2nd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ♦ If short circuit to B+, then all hydraulic control elements are switched off

► Breakdown

P17AA

Diagnostic information - PDK control unit

Electric pressure adjuster 2 (clutch 2)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 2 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 2 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Pressure adjuster 2 not energized
- Momentary current of low-side pressure adjuster 2 = 0 mA
- Momentary current of low-side pressure adjuster 2 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 2 is switched off
- ♦ Low-side electric pressure adjuster 2 is switched off
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ The maximum cooling activation pressure is increased to prevent activation of the emergency stop function

- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- ▶ Breakdown

P17AB

Diagnostic information - PDK control unit

Electric pressure adjuster 3 (system pressure)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 3 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 3 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Pressure adjuster 3 not energized
- Momentary current of low-side pressure adjuster 3 = 0 mA
- Momentary current of low-side pressure adjuster 2 = 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 2 is switched off
- ♦ Low-side electric pressure adjuster 2 is switched off
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No limp-home program diagnostics
- ♦ The maximum cooling activation pressure is increased to prevent activation of the emergency stop function

- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- ▶ Breakdown

P17AC

Diagnostic information - PDK control unit

Electric pressure adjuster 4 (cooling flow)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 4 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 4 = 0 mA
- Momentary current of low-side pressure adjuster 4 $\geq$ 850 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side electric pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift

rod

- ◆ The maximum system pressure is increased to prevent activation of the emergency stop function
- ◆ Pressure-sensor plausibility check is switched off
- ◆ If short circuit to B+, then all hydraulic control elements are switched off
- Breakdown

P17AD

Diagnostic information - PDK control unit

Electric pressure adjuster 5 (synchro selection 6, R, 3, 5)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 5 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary current of low-side pressure adjuster 5 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side pressure adjuster 5 = 0 mA

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop mode
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop

function

- ◆ The last engaged gear is held

P17AE

Diagnostic information - PDK control unit

Electric pressure adjuster 6 (synchro selection 4, 2, 1, 7)

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 6 faulty

Fault finding information

- If the fault is still statically present after all possible fault causes have been checked and cannot be deleted, the PDK control unit might be faulty and have to be replaced.

Fault setting condition(s)

- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Hydraulic control 3 is switched off
- ♦ Low-side pressure adjusters 4, 5, 6 are switched off
- ♦ Low-side solenoid valves 1, 2 are switched off
- ♦ Gear held
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Top speed limitation via engine intervention (lowest engaged gear)
- ♦ One-touch operation and Sport program are deactivated
- ♦ All adaptations are deactivated
- ♦ Pressure adjusters 5 and 6 are depressurized, and multiplexer is changed over to locked shift rod
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function

♦ Last engaged gear is held

Diagnosis information PDK control unit

Electrical pressure adjuster 7

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to ground
- ♦ Short circuit to B+
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Pressure adjuster 7 faulty

Fault setting condition(s)

- Momentary current of low-side electrical pressure adjuster 6 = 0 mA
- Momentary pressure regulator current < setpoint current (maximum deviation 35 mA)
- Momentary pressure regulator current > setpoint current (maximum deviation 35 mA)
- Momentary current of low-side electrical pressure adjuster 6 = 850 mA
- Momentary current of low-side electrical pressure adjuster 6 = 0 mA

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Controlled transverse-differential lock is deactivated
- ♦ Hydraulic control 4 is switched off
- ♦ Low-side electrical pressure adjuster 7 is switched off
- ♦ Shiftlock solenoid is deactivated

P17B1

Diagnosis information PDK control unit

Clutch 1 - activation pressure implausible

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- Comparison of the current clutch pressure with the current touchpoint + an offset (1 bar)
- Comparison of the current clutch pressure with the prefilling pressure + an offset (0.6 bar)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 2 (only 2nd, 4th, 6th gears possible)
- ♦ Low-side electrical pressure adjuster 1 is switched off
- ♦ No R gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ All adaptations are deactivated
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P17B2

Diagnosis information PDK control unit

Clutch 2 - activation pressure implausible

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- Comparison of the current clutch pressure with the current touchpoint + an offset (1 bar)
- Comparison of the current clutch pressure with the prefilling pressure + an offset (0.6 bar)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 1 (only 1st, 3rd, 5th, 7th and R gears possible)
- ♦ Low-side electrical pressure adjuster 2 is switched off
- ♦ Moving off in all gears
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ All adaptations are deactivated
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P17B3

Diagnosis information PDK control unit

Clutch valve 1 remains mechanically at excessively high pressure at overlap shifting after rapid filling

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- Pressure in target gear > filling pressure + offset (10 bar)
- Prefiltering complete

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 2 (only 2nd, 4th, 6th gears possible)
- ♦ Clutches 1 and 2 are opened
- ♦ All gears in the transmission are set to neutral
- ♦ Low-side electrical pressure adjuster 1 is switched off
- ♦ Moving off in all gears
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ All adaptations are deactivated
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P17B4

Diagnosis information PDK control unit

Clutch valve 2 remains mechanically at excessively high pressure at overlap shifting after rapid filling

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- Pressure in target gear > filling pressure + offset (10 bar)
- Prefiltering complete

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 1 (only 1st, 3rd, 5th, 7th and R gears possible)
- ♦ Clutches 1 and 2 are opened
- ♦ All gears in the transmission are set to neutral
- ♦ Low-side electrical pressure adjuster 2 is switched off
- ♦ Moving off in all gears
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ All adaptations are deactivated
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P17B5

Diagnosis information PDK control unit

Clutch 1 slips at low, high differential speed

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- Engine speed - shaft speed 1 - desired slip > maximum permissible slip
- Engine speed - shaft speed 1 - desired slip > maximum permissible slip (300 rpm)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 2 (only 2nd, 4th, 6th gears possible)
- ♦ No R gear
- ♦ Moving off in 2nd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ All adaptations are deactivated
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P17B6

Diagnosis information PDK control unit

Clutch 2 slips at low, high differential speed

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- Engine speed - shaft speed 2 - desired slip > maximum permissible slip
- Engine speed - shaft speed 2 - desired slip > maximum permissible slip (300 rpm)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 1 (only 1st, 3rd, 5th, 7th and R gears possible)
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ All adaptations are deactivated
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P17B7

Diagnosis information PDK control unit

Clutch 1 slips at low, high differential speed

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- Engine speed - shaft speed 1 - desired slip > maximum permissible slip
- Engine speed - shaft speed 1 - desired slip > maximum permissible slip (300 rpm)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 2 (only 2nd, 4th, 6th gears possible)
- ♦ No R gear
- ♦ Moving off in 2nd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ All adaptations are deactivated
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P17B8

Diagnosis information PDK control unit

Clutch 2 slips at low, high differential speed

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- Engine speed - shaft speed 2 - desired slip > maximum permissible slip
- Engine speed - shaft speed 2 - desired slip > maximum permissible slip (300 rpm)

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 1 (only 1st, 3rd, 5th, 7th and R gears possible)
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ All adaptations are deactivated
- ♦ The maximum system pressure is increased to prevent activation of the emergency stop function
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P17BB

Diagnosis information PDK control unit

Clutch 1 cannot be opened

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- After the timer runs down (clutch push-out time), the current pressure is compared with the current touchpoint + an offset

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 2 (only 2nd, 4th, 6th gears possible)
- ♦ Low-side electrical pressure adjuster 1 is switched off
- ♦ Moving off in all gears
- ♦ No R gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ All adaptations are deactivated
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

P17BC

Diagnosis information PDK control unit

Clutch 2 cannot be opened

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- After the timer runs down (clutch push-out time), the current pressure is compared with the current touchpoint + an offset

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Driving with gear train half 1 (only 1st, 3rd, 5th, 7th and R gears possible)
- ♦ Low-side electrical pressure adjuster 2 is switched off
- ♦ Moving off in all gears
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ All adaptations are deactivated
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

Diagnosis information PDK control unit

Clutch 1 cannot be opened during shift overlap

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- Overrun upshift or traction downshift is checked prior to overlap
 - Traction upshift and overrun downshift are checked after overlap
 - Detection of the type of shifting
 - Timer; shifting takes place when the state is reached.
 - The following must take place within a time of 1.5 - 3 s:
 - difference between engine speed and target speed must be <20%
- or
- 5 time intervals with a time base of 30 ms must reach a certain quotient

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Gear is held
- ♦ Low-side electrical pressure adjuster 2 is switched off
- ♦ Only 1st, 3rd, 5th, 7th and R gears are possible
- ♦ Moving off in 3rd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ One-touch operation and Sport program are deactivated
- ♦ No rpm control
- ♦ No positive engine intervention
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.
- ♦ Hotmode display is activated

♦ Engine stalls when vehicle stops

Diagnosis information PDK control unit

Clutch 2 cannot be opened during shift overlap

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Open circuit
- ♦ Short circuit to B+
- ♦ Short circuit to ground
- ♦ Loose contact or corrosion on connectors or lines
- ♦ Clutch faulty
- ♦ Hydraulics faulty

Fault setting condition(s)

- Overrun upshift or traction downshift is checked prior to overlap
- Traction upshift and overrun downshift are checked after overlap
- Detection of the type of shifting
- Timer; shifting takes place when the state is reached.
- The following must take place within a time of 1.5 - 3 s:
 - difference between engine speed and target speed must be <20%
- or
- 5 time intervals with a time base of 30 ms must reach a certain quotient

Fault effects

- ♦ Transmission limp-home display in the instrument cluster control unit
- ♦ Gear is held
- ♦ Low-side electrical pressure adjuster 1 is switched off
- ♦ Only 2nd, 4th and 6th gears are possible
- ♦ No R gear
- ♦ Moving off in 2nd gear
- ♦ No start/stop
- ♦ Torque limitation
- ♦ No emergency program diagnostics
- ♦ One-touch operation and Sport program are deactivated
- ♦ No rpm control
- ♦ No positive engine intervention
- ♦ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.

◆ Hotmode display is activated

P17E0

Diagnosis information PDK control unit

Shift valve stuck

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ Leakage in hydraulics
- ♦ Shift valve faulty (mechanical blockage)
- ♦ Hydraulic control unit faulty

Fault setting condition(s)

- The test is performed only once during the ignition cycle
- The test is performed the first time the pressure retaining valve is set (with preloading of the emergency concept). This prevents unintentional activation of the hydraulic emergency concept

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit
- ♦ Torque limitation
- ♦ Low-side electric pressure adjuster 3 is switched off
- ♦ No limp-home program diagnostics
- ♦ No pressure sensor plausibility check
- ♦ Clutch pressure < holding threshold (8.1 bar)

P17F1

Diagnosis information PDK control unit

Transmission overtemperature, stage 2

Information

For this purpose, observe the Technical Information for model years 2013 and 2014: **3734 - PDK transmission: fault codes P172D, P17F1, P17F2, P18CA, P18CB (65/13)**

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ At high outside temperatures in combination with a very sporty driving style
- ♦ Heat exchanger does not work
- ♦ Transmission temperature sensor faulty

Fault setting condition(s)

- Oil sump temperature > 145 °C (293 °F)
- Calculated steel plate temperature > temperature threshold over time (690 - 300 °C/1,274 - 572 °F)

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit hotmode display 2
- ♦ Torque limitation

Fault finding notes

- ♦ Allow transmission to cool down
- ♦ Erase fault memory and carry out a test drive

P17F2

Diagnosis information PDK control unit

Transmission overtemperature, stage 3

Information

For this purpose, observe the Technical Information for model years 2013 and 2014: **3734 - PDK transmission: fault codes P172D, P17F1, P17F2, P18CA, P18CB (65/13)**

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ At high outside temperatures in combination with a very sporty driving style
- ♦ Heat exchanger does not work
- ♦ Transmission temperature sensor faulty

Fault setting condition(s)

- Oil pan temperature > 150 °C (302 °F) or
- Calculated steel plate temperature > temperature threshold over time (691 - 301 °C/1,276 - 574 °F) or
- Spray oil evaluation > 500 °C (932 °F)

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit hotmode display 3
- ♦ Clutches 1 and 2 are opened

Fault finding notes

- ♦ Allow transmission to cool down
- ♦ Erase fault memory and carry out a test drive

P17F5

Diagnosis information PDK control unit

Transmission overtemperature, stage 1

Diagnostic conditions

- Driving in all gears

Possible fault causes

- ♦ At high outside temperatures in combination with a very sporty driving style
- ♦ Heat exchanger does not work
- ♦ Transmission temperature sensor faulty

Fault setting condition(s)

- Gear wheel set temperature (calculated) > 145 °C (293 °F)

Fault effects

- ♦ Transmission limp-home display in instrument cluster control unit hotmode display 1
- ♦ Torque limitation

Fault finding notes

- ♦ Allow transmission to cool down
- ♦ Erase fault memory and carry out a test drive

P181A

Diagnosis information PDK control unit

**Pressure sensor 1, electrical fault
or
shadow fault**

INFORMATION

- ♦ From **software version Q033**, an electrical fault in pressure sensor 1 is indicated only via fault code **P0843**.
- ♦ From this software version, an operating state that could have been caused by brief sticking of one of the pressure adjusters is documented with fault entry **P181A**.
⇒ This leads to flushing of the pressure adjusters during driving, and this can cause one-time shift delays. However, there are no further impairments for the driver.

 There is generally no fault in this case.

♦ **Whether an electrical fault was present** can be read in the extended fault memory under > **Electrical fault if value = 1** <.

- If the value is **1**, an electrical fault was detected in pressure sensor 1.
 - ▶ Perform fault finding.
- There was no electrical fault if the value is **0**; only flushing was performed.
 - ▶ Erase fault memory.

Diagnostic conditions

- Ignition on
- Trip with different operating conditions

Possible fault causes

- **For P181A** ⇒ see information above
- **For P0843:**
 - ♦ Open circuit
 - ♦ Short circuit to B+
 - ♦ Short circuit to ground
 - ♦ Loose contact or corrosion on connectors or lines
 - ♦ Pressure sensor 1 faulty

Fault setting condition(s)

- Measured pressure > threshold value for short circuit to B+ (21 bar)
- Measured pressure < threshold value for short circuit to ground or open circuit (0.35 bar)

- If the faults relating to the oil temperature, pressure sensor 1 or pressure sensor 2 are entered, then the sensor ground is missing

Fault effects

- ◆ Transmission limp-home display in the instrument cluster control unit
- ◆ Driving with gear train half 2 (only 2nd, 4th and 6th gears)
- ◆ No R gear
- ◆ Moving off in all gears
- ◆ No start/stop
- ◆ Torque limitation
- ◆ No limp-home program diagnostics
- ◆ Deactivate all adaptations
- ◆ Clutch-overlap function is deactivated, i.e. one clutch is not closed until the other clutch is opened during the gear-changing process.